Cross-frequency coupling

Instructor: Mark Kramer

Today

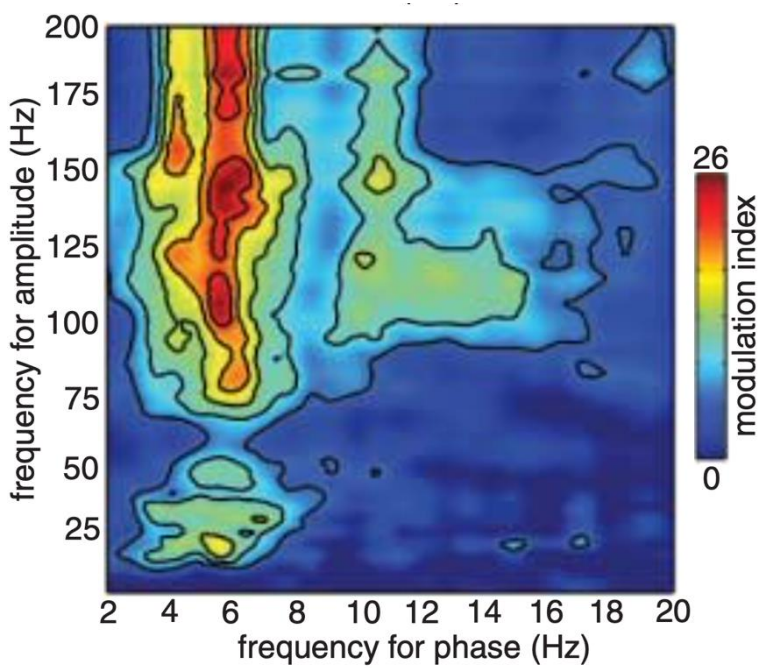
Cross-frequency coupling (one type of)

Remember, coherence

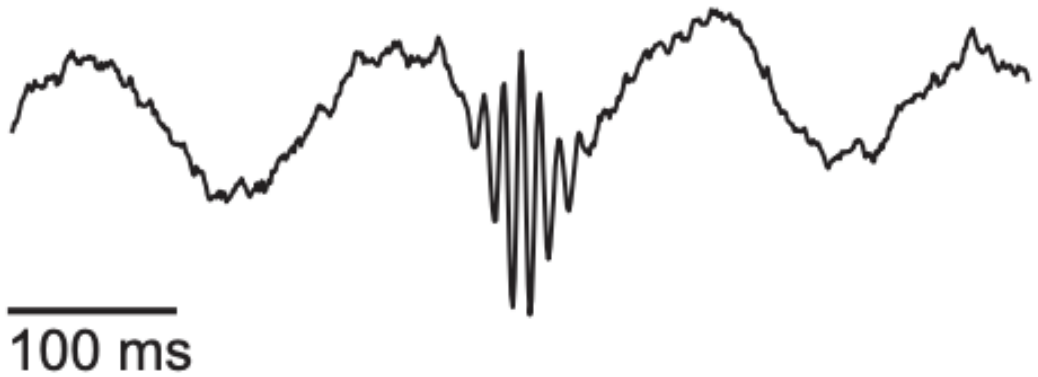
a constant phase relationship between two signals across trials,
at the same frequency

Today, coupling **between different frequencies**

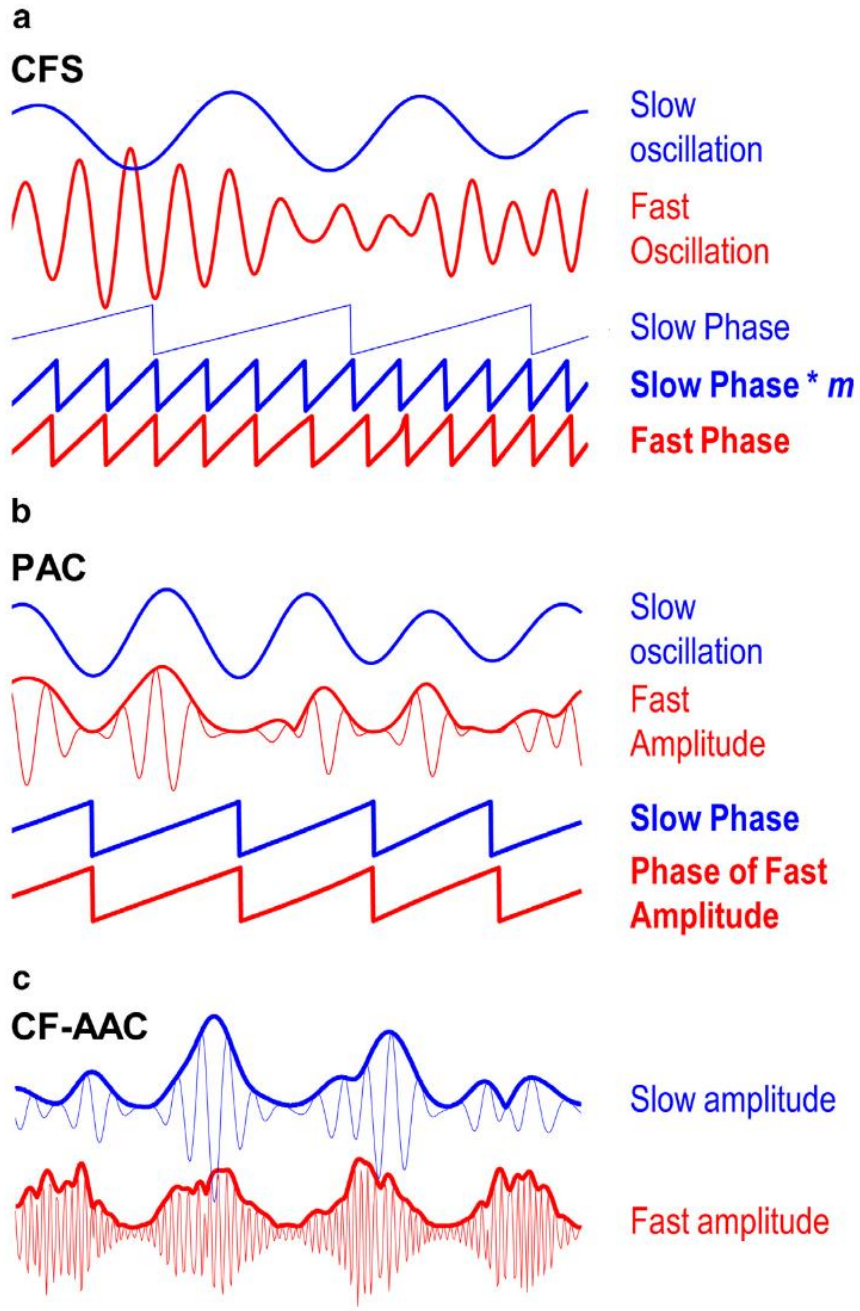
Examples



[Canolty et. al., Science, 2006]



[Tort et. al., PNAS, 2008]

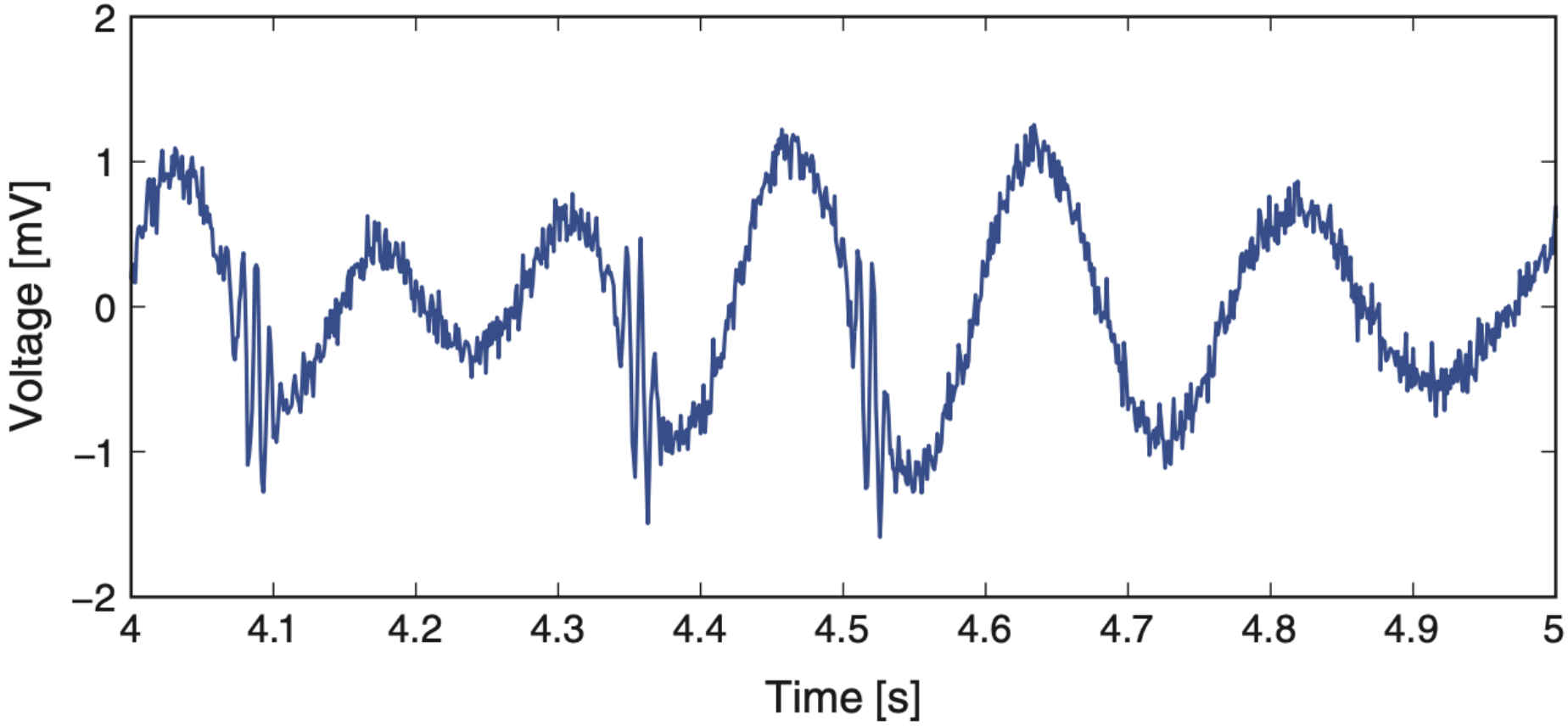


[Palva & Palva, EJN, 2018]

Outline

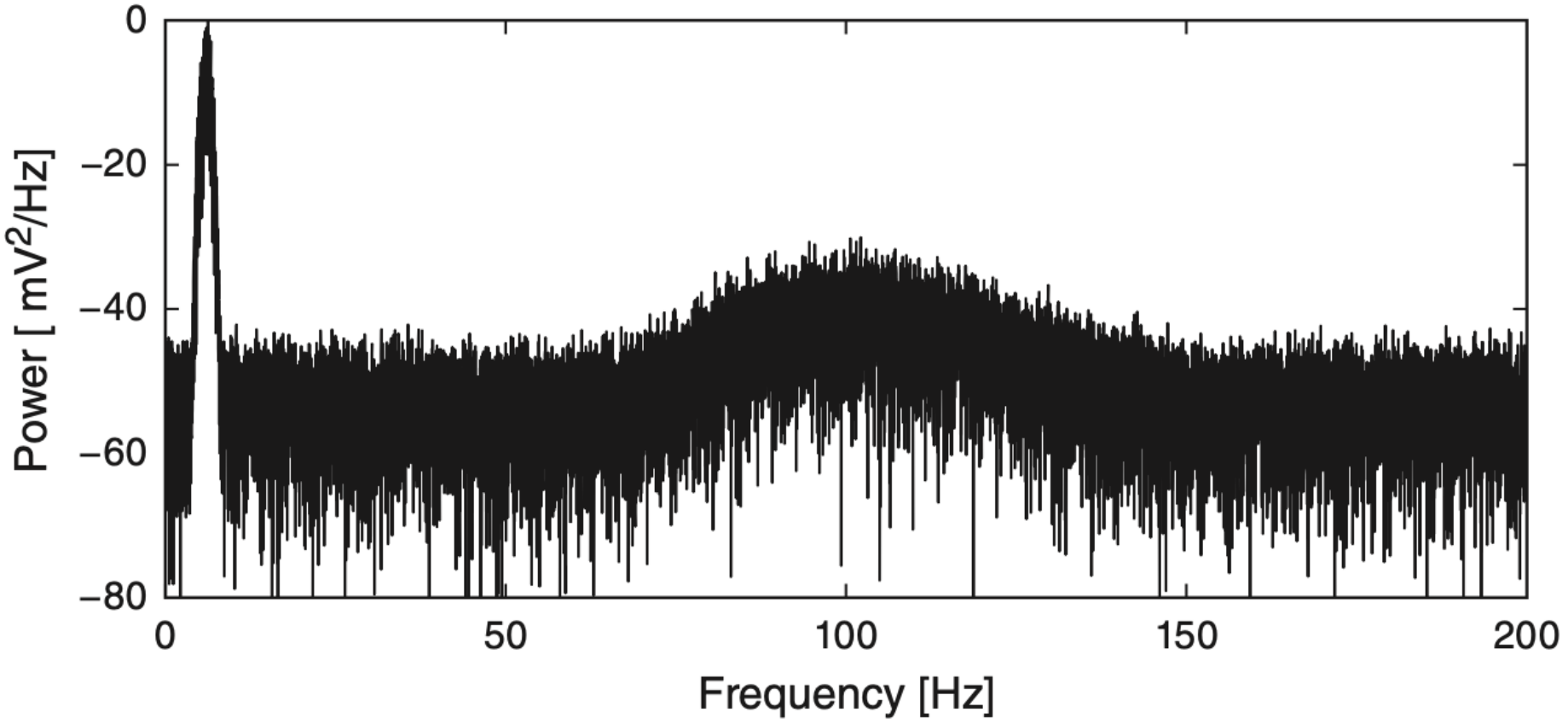
Data	100 s of local field potential data sampled at 1000 Hz.
Goal	Characterize the coupling between rhythms of different frequency.
Tools	Hilbert transform, analytic signal, instantaneous phase, cross-frequency coupling.

Data



Q. How to make sense of these data?

Spectrum



Q. What do you see?

CFC in three steps

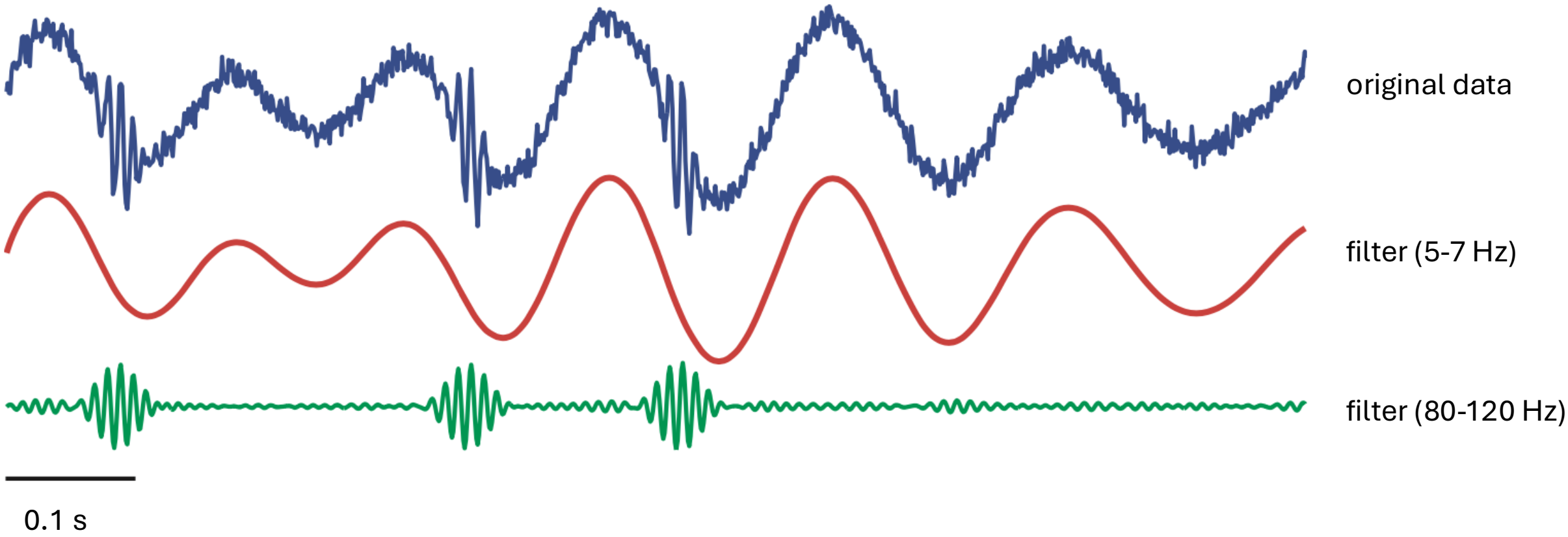
CFC analysis steps

1. Filter the data into high- and low-frequency bands.
2. Extract the amplitude and phase from the filtered signals.
3. Determine if the phase and amplitude are related.

Let’s perform each step …

CFC – Step 1

Filter the Data into High- and Low-Frequency Bands

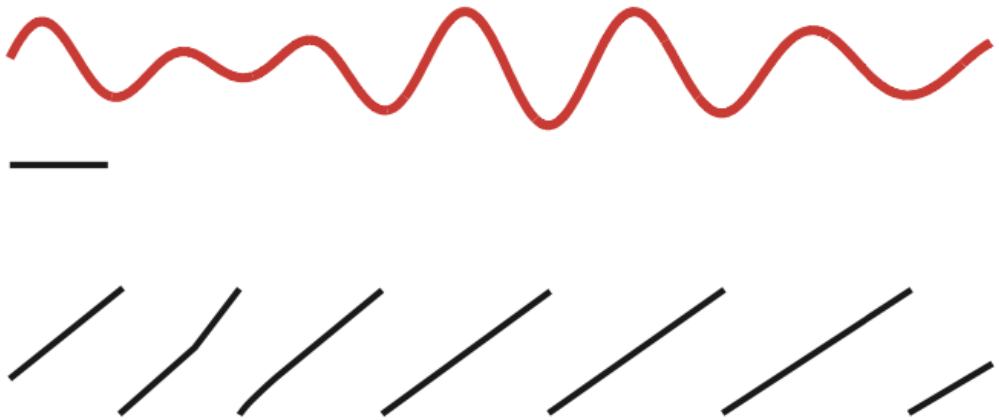


Q. Why did we filter in these bands?

CFC – Step 2

Extract the Amplitude and Phase from Filtered Signals

filter (5-7 Hz)



phase (of low frequency band)

filter (80-120 Hz)



amplitude envelope (of high frequency band)

Q. How?

Hilbert transform

$$y = H(x)$$

$$H(x) = \begin{cases} -\pi/2 \text{ phase shift} & \text{if } f > 0, \\ 0 \text{ phase shift} & \text{if } f = 0, \\ \pi/2 \text{ phase shift} & \text{if } f < 0. \end{cases}$$

The Hilbert transform $H(x)$ of the signal x produces a phase shift of ± 90 degrees for $\mp$ frequencies of x.

Hilbert transform

Define: Analytic signal z

$$z = x + iy = x + iH(x)$$

Impact: remove negative frequencies from z

Q. How?

Hilbert transform

Q. What does it do?

Ex.

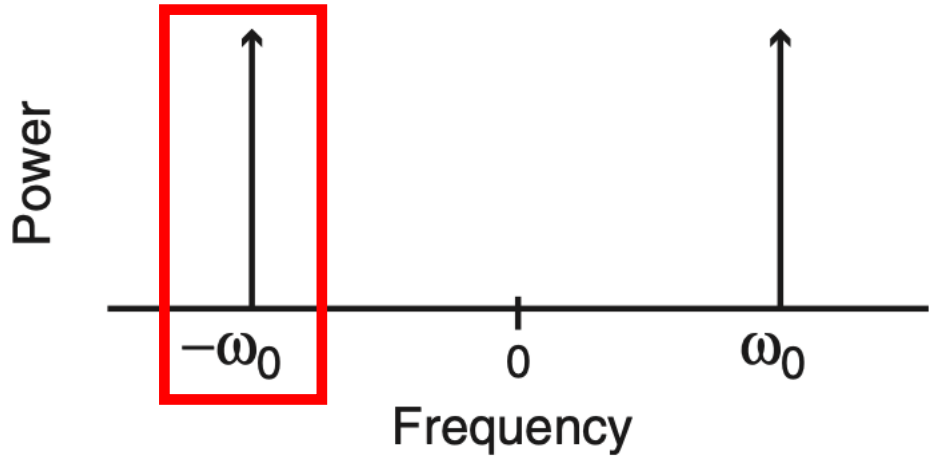
$$x_0 = 2\cos(2\pi f_o t) = 2\cos(\omega_0 t) \quad \text{where} \quad \omega_0 = 2\pi f_o$$

Euler's formula

$$x_0 = e^{i\omega_0 t} + e^{-i\omega_0 t}$$

positive frequency negative frequency

we usually ignore this one



Note: The spectrum has two peaks

Hilbert transform

Apply the Hilbert transform to x_0.

$$x_0 = e^{i\omega_0 t} + e^{-i\omega_0 t}$$

$$H(x) = \begin{cases} -\pi/2 \text{ phase shift} & \text{if } f > 0, \end{cases}$$

➔ multiply positive frequency part of x by $-i$

Q. Really?

Consider $e^{i\omega_0 t}$

positive frequency part of x

Shift $e^{i\omega_0 t}$ by $-\frac{\pi}{2}$

➔ $e^{i(\omega_0 t - \frac{\pi}{2})}$ ➔ $e^{i\omega_0 t} e^{-i\pi/2}$ ➔ $e^{i\omega_0 t}(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2})$ ➔ $e^{i\omega_0 t}(-i)$

Hilbert transform

Apply the Hilbert transform to x_0.

$$x_0 = e^{i\omega_0 t} + e^{-i\omega_0 t}$$

$$H(x) = \begin{cases} -\pi/2 \text{ phase shift} & \text{if } f > 0, \\ 0 \text{ phase shift} & \text{if } f = 0, \\ \pi/2 \text{ phase shift} & \text{if } f < 0. \end{cases} \quad \rightarrow \quad H(x) = \begin{cases} -ix & \text{if } f > 0, \\ x & \text{if } f = 0, \\ ix & \text{if } f < 0. \end{cases}$$

So, $x_0 = e^{i\omega_0 t} + e^{-i\omega_0 t}$

multiply by $-i$ multiply by i

$$y_0 = H(x_0) = -ie^{i\omega_0 t} + ie^{-i\omega_0 t}$$

Euler's formula

$$= 2\sin(\omega_0 t)$$

Hilbert Transform of x_0 (a cosine function) is a sine function.

Hilbert transform

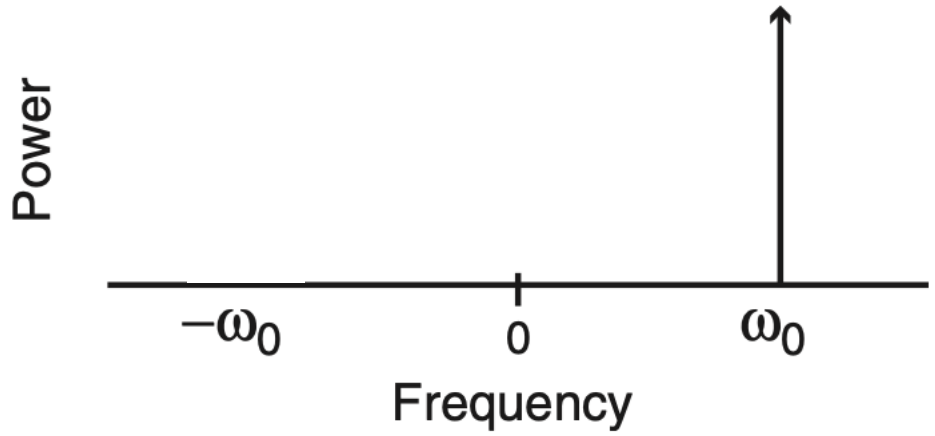
Analytic signal z

$$z = x + iy = x + iH(x)$$

$$2\cos(\omega_0 t) \qquad 2\sin(\omega_0 t)$$

$$= 2\cos(\omega_0 t) + i2\sin(\omega_0 t)$$

$$\boxed{= 2e^{i\omega_0 t}}$$



The analytic signal contains
no negative frequencies

Hilbert transform

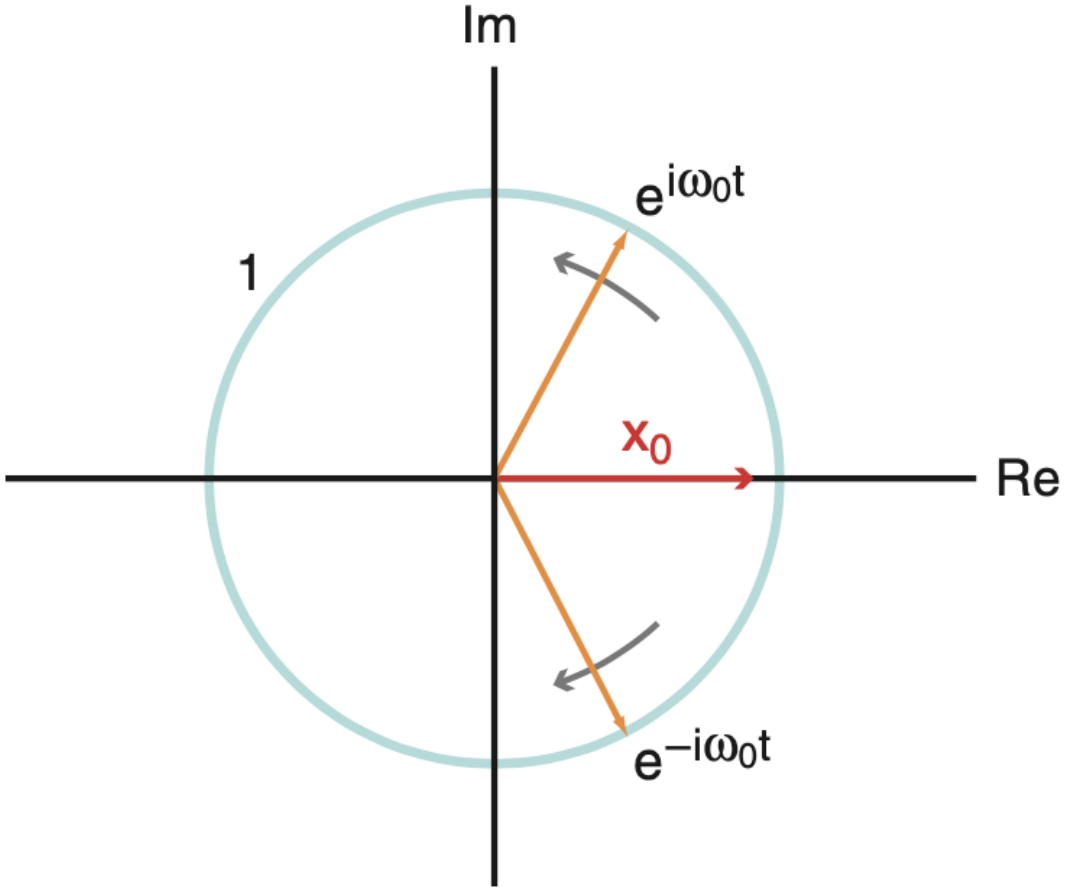
Original signal $x_0 = 2\cos(2\pi f_o t) = e^{i\omega_0 t} + e^{-i\omega_0 t}$

Complicated (2 complex exponentials)

Analytic signal $z_0 = 2e^{i\omega_0 t}$

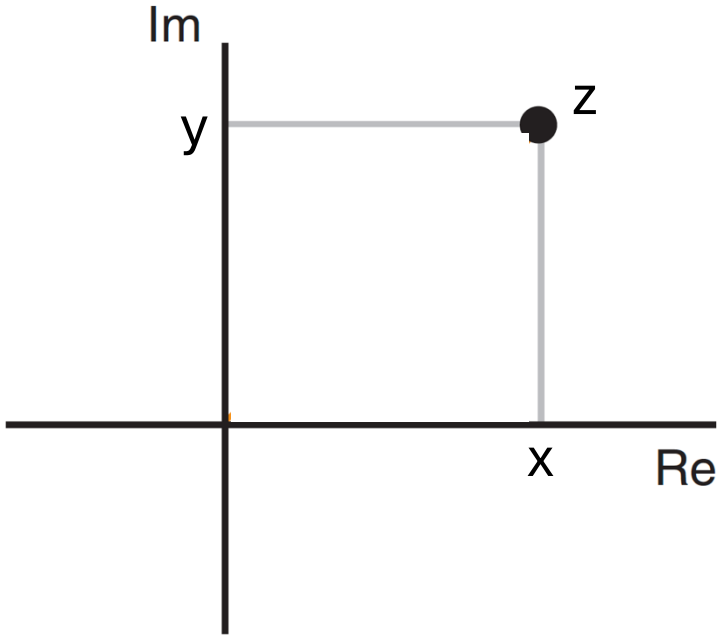
Simple (1 complex exp)

A point in the complex plane



Hilbert transform

Analytic signal $z = x + iy$ A point in the complex plane



$$z(t) = A(t)\, e^{i\,\phi(t)}$$

amplitude phase

Get the **amplitude** and **phase** from the analytic signal

Ex.

$z_0(t) = 2e^{i\omega_0 t}$ $A(t) = 2$ $\phi(t) = \omega_0\, t$

CFC in three steps

CFC analysis steps

1. Filter the data into high- and low-frequency bands.
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CFC – Step 3

Determine if the Phase and Amplitude are Related

Define the two-column vector

$$\begin{pmatrix} \phi(1) & A(1) \\ \phi(2) & A(2) \\ \phi(3) & A(3) \\ \vdots & \vdots \end{pmatrix}$$



<u>phase</u> of low frequency band activity

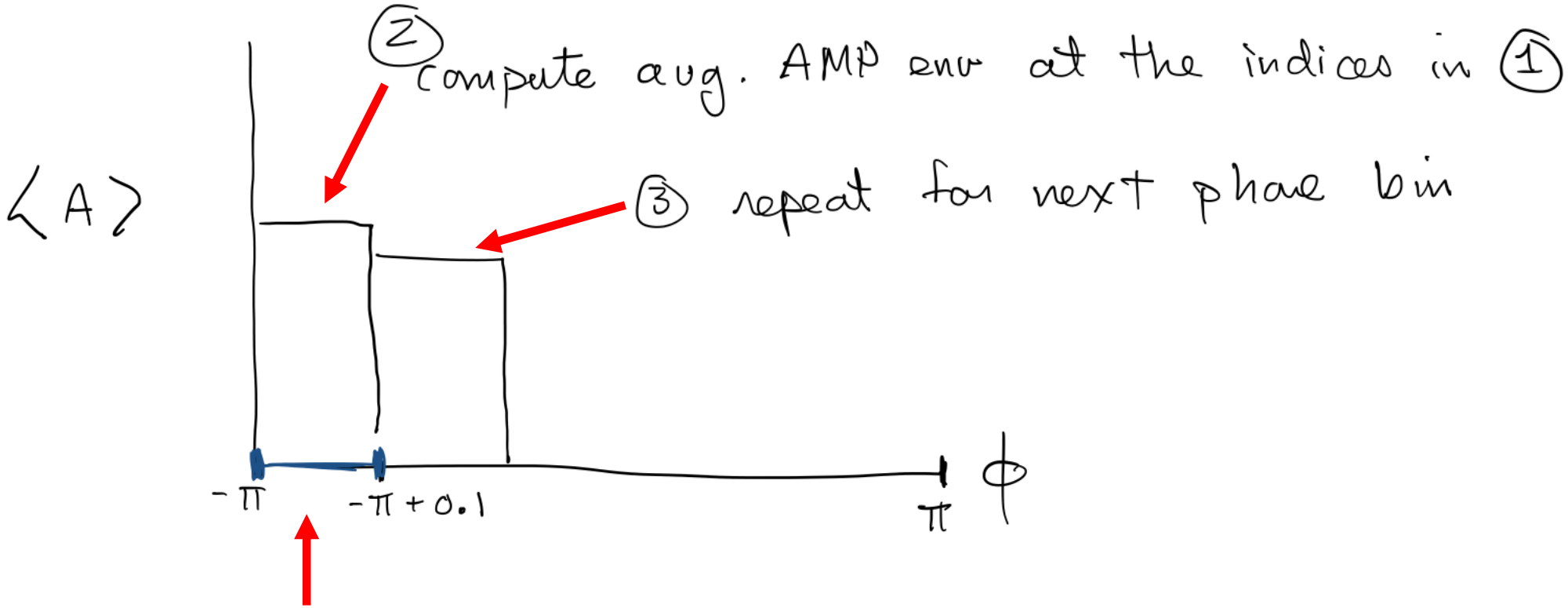
<u>amplitude</u> of high frequency band activity

Make a histogram

CFC – Step 3

Determine if the Phase and Amplitude are Related

Divide the phase into bins of size 0.1

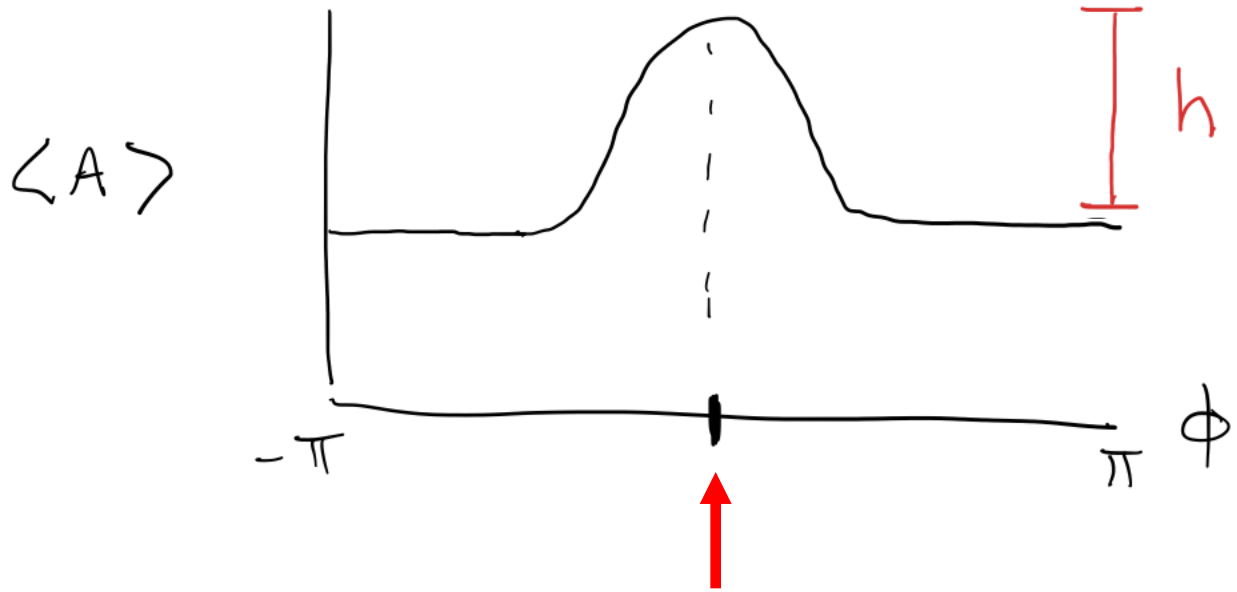


(1) Find all ϕ in this range $[-\pi, -\pi + 0.1]$
& get indices $\{1, 2, 3, 10, 11, 12, 41, 42, 43, \ldots\}$

CFC – Step 3

Determine if the Phase and Amplitude are Related

If phase modulate amplitude



summarize extent of modulation

Q. What does no phase-amplitude coupling look like in the plot?

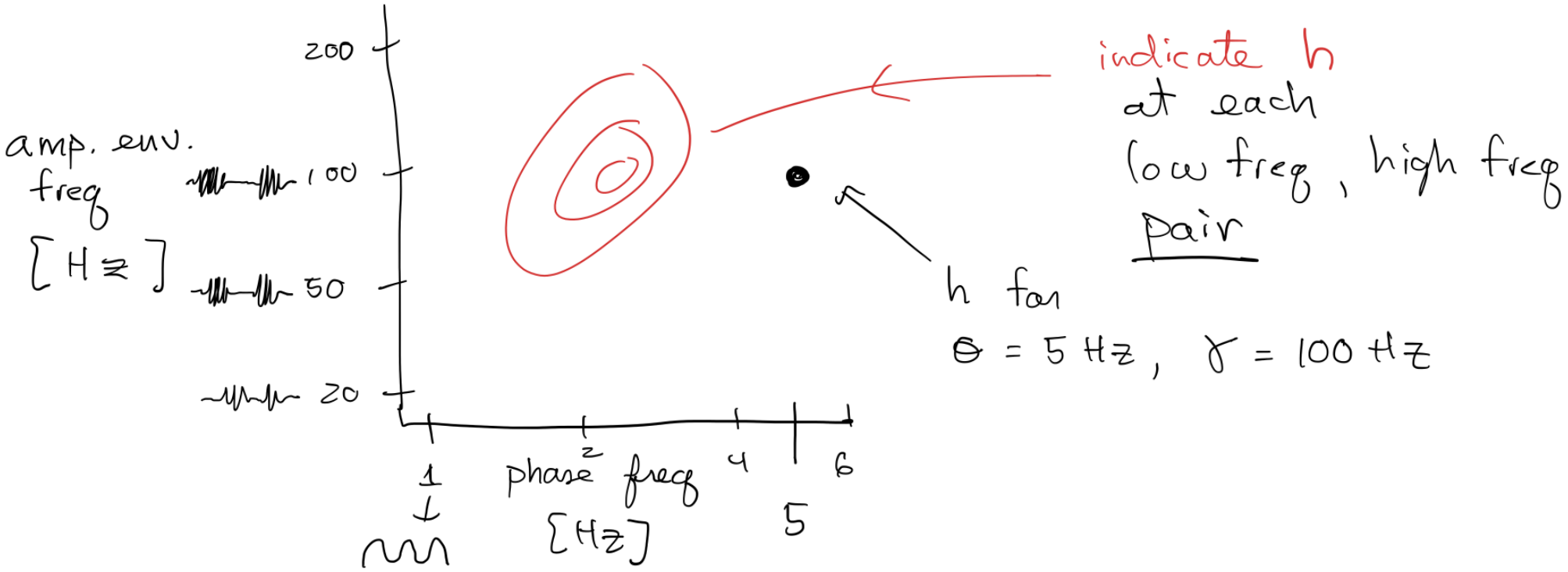
At this phase, amplitude envelope is big

➔ phase modulates amplitude

CFC – Step 4

(optional): Repeat for other frequencies.

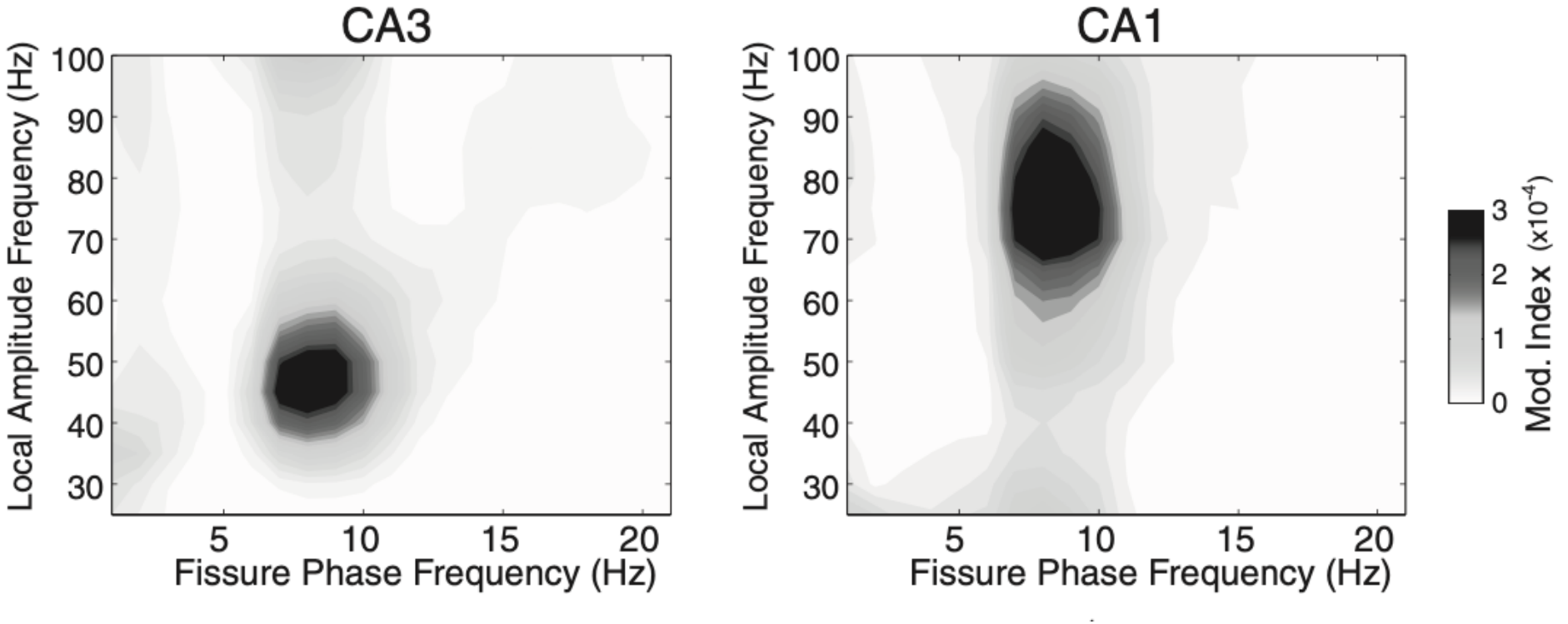
Summarize in a <u>comodulogram</u>



CFC – Step 4

(optional): Repeat for other frequencies.

Summarize in a <u>comodulogram</u>



[Tort et al., J Neurophysiol, 2010]

CFC in three steps

CFC analysis steps

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Python